Finpro Bank

Data Architecture, Implementation and Recommendations

Mohammad Shahidullah

Outline

- Introduction
- Data acquisition and cleaning
- Database design and setup
- Evaluation and interpretation
- Data integration and security
- Data reporting
- General handover documentation
- Insights and recommendations
- Summary
- Appendix

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Introduction

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Purpose of the presentation

- Present a complete, structured handover of the FinPro Bank data management system I designed and implemented.
- Document the full lifecycle of activities—from data acquisition and cleaning to reporting and dashboard development.
- Ensure the incoming Data Manager can seamlessly maintain, troubleshoot, and enhance the existing data architecture.
- Provide clear, human-centered explanations, supported by screenshots, to make the transition smooth and efficient.

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Summary of work done on the project

- Performed end-to-end data acquisition, cleaning, validation, and transformation across multiple banking datasets.
- Designed and implemented a relational database schema, including tables, constraints, indexing, and optimized storage.
- Built structured ETL workflows to integrate customer, branch, transaction, and geographic data into a unified system.
- Applied security and access controls, including user roles, privilege segmentation, and encryption of sensitive fields.
- Developed interactive Tableau worksheets, dashboards, and storylines for branch-level, country-level, and overall performance analysis.
- Prepared comprehensive handover documentation, ensuring clarity, maintainability, and continuity for the next Data Manager.

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Data acquisition and cleaning

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Data cleaning documentation (1 of 3)

Data Sources Used

- **Customer_Data.csv**
  Contains customer demographics, age, gender, branch association, and account details.
- **Transaction_Data.csv**
  Includes all financial transactions with timestamps, amounts, transaction types, and customer IDs.
- **Branch_Data.csv**
  Provides branch codes, branch names, and geographic attributes.
- **US_Cities.csv / Country Mapping File**
  Used for geolocation, country-level mapping, and visualization enrichment.

Purpose of These Sources

- Together, these datasets form the foundation for customer analytics, branch performance evaluation, and country-wise transaction insights.

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Data cleaning documentation (2 of 3)

Data Cleaning Steps Implemented and Data Cleaning Workflow

- **Handled Missing Values**
  - Filled missing branch codes using customer-branch mapping.
  - Removed rows with critical missing identifiers (CustomerID, TransactionID).

- **Removed Duplicates**
  - Identified duplicate customer records using CustomerID.
  - Removed repeated transaction entries based on TransactionID + Timestamp.

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Data cleaning documentation (3 of 3)

- **Standardized Data Formats**
  - Converted dates to ISO format.
  - Normalized text fields (branch names, country names).
  - Standardized currency values.

- **Outlier Treatment**
  - Validated negative balances and corrected data entry errors.

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Analysis and visualization using MS Excel (1 of 2)

Excel Formulas Used

- Duplicate Detection:
  `=COUNTIF(A:A, A2)`
- Missing Value Checks:
  `=IF(ISBLANK(A2), "Missing", "OK")`
- Age Calculation from Date of Birth:
  `=DATEDIF([@DateOfBirth], TODAY(), "Y")`
- For new column TransactionMonth:
  `=TEXT([@TransactionDate], "MMMM")`

VBA macro to

- Remove duplicates
- Standardize date formats
- Apply conditional formatting for anomalies

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Analysis and visualization using MS Excel (2 of 2)

Data Cleaning Principles Applied

- Accuracy
- Consistency
- Completeness
- Integrity
- Auditability

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Database design and setup

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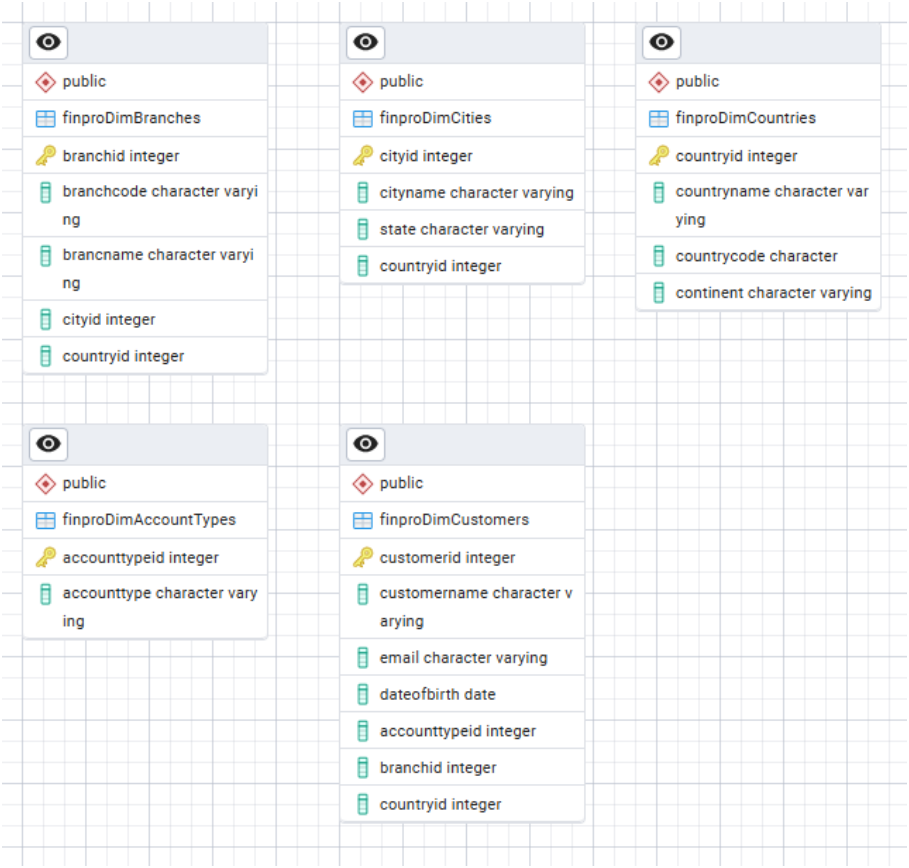
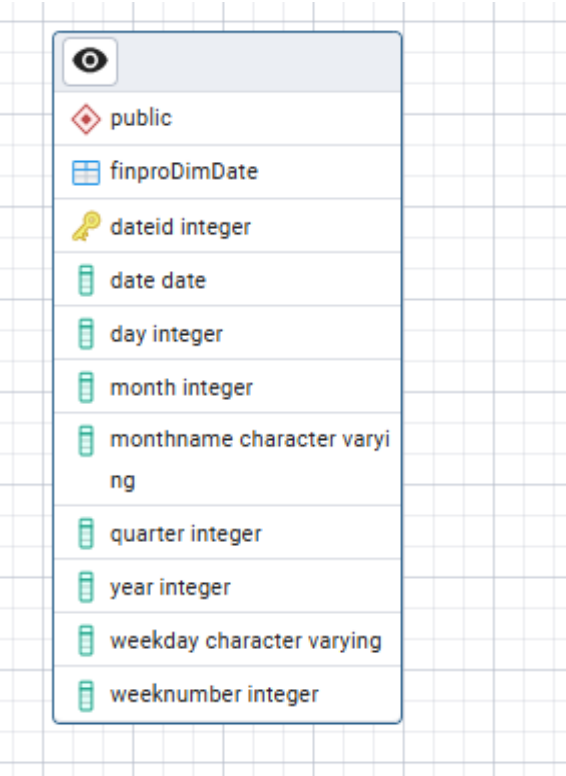
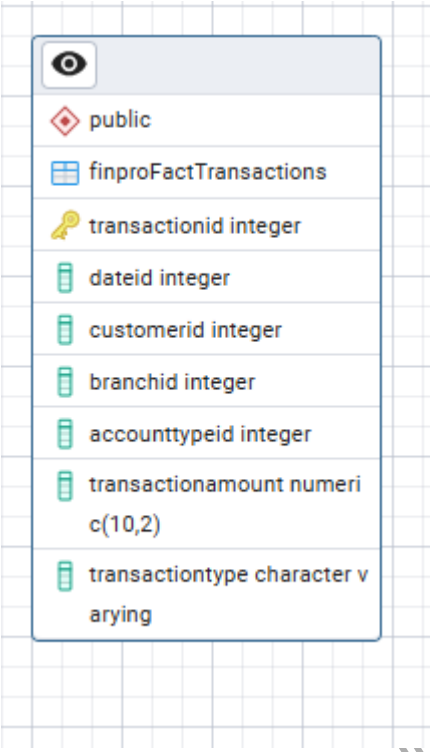
Entity relationship diagram (ERD) (1 of 3)



Dimension Tables



Dimension Date Table



Central Fact Table

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Entity relationship diagram (ERD) (2 of 3)

Dimension Tables

- finproDimCustomers
  Stores customer demographics, DOB, account type, and branch association.
- finproDimBranches
  Contains branch codes, names, and geographic identifiers.
- finproDimCities
  Provides city-level attributes including state and country mapping.
- finproDimCountries
  Stores country names, codes, and continent information.
- finproDimAccountTypes
  Defines account categories such as Savings, Checking, etc.
- finproDimDate
  A full date dimension enabling year, quarter, month, week, and day-level analysis.

Fact Table

- finproFactTransactions
  Captures all financial transactions with links to customer, branch, account type, and date.

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Entity relationship diagram (ERD) (3 of 3)

Key Relationships

- Customers → Branches (Many-to-One)
  Each customer belongs to a single branch.
- Branches → Cities → Countries (Hierarchical)
  Enables geographic roll-ups for reporting.
- Transactions → Customers / Branches / Account Types / Date
  The fact table references all dimensions to support multi-angle analysis.

Business Logic Behind the Design

- Ensures **single source of truth** for customer, branch, and geographic attributes.
- Supports **time-series analysis** through the date dimension.
- Enables **drill-down reporting** (Country → City → Branch → Customer → Transaction).
- Follows best practices in **data warehousing** for performance, scalability, and clarity.

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Table structure design

Table	Key Columns	Data Types	Constraints
finproDimCustomers	customerid (PK), customername, email, dateofbirth, accounttypeid, branchid, countryid	int, varchar, date	PK(customerid), FK(accounttypeid), FK(branchid), FK(countryid)
finproDimBranches	branchid (PK), branchcode, branchname, cityid, countryid	int, varchar	PK(branchid), FK(cityid), FK(countryid)
finproDimCities	cityid (PK), cityname, state, countryid	int, varchar	PK(cityid), FK(countryid)
finproDimCountries	countryid (PK), countryname, countrycode, continent	int, varchar	PK(countryid)
finproDimAccountTypes	accounttypeid (PK), accounttype	int, varchar	PK(accounttypeid)
finproDimDate	dateid (PK), date, day, month, monthname, quarter, year, weekday, weeknumber	int, date, varchar	PK(dateid)
finproFactTransactions	transactionid (PK), dateid, customerid, branchid, accounttypeid, transactionamount, transactiontype	int, numeric, varchar	PK(transactionid), multiple FKs

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OLAP schema (1 of 3)

Schema Used: Snowflake Schema

- The FinPro Bank data model follows a snowflake schema, where dimension tables are normalized into multiple related tables (e.g., Branch → City → Country).

Why Snowflake?

- Reduces redundancy
- Improves data integrity
- Supports hierarchical reporting (Country → City → Branch)
- Aligns with enterprise-grade data warehousing standards

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OLAP schema (2 of 3)

Fact Table

- **finproFactTransactions**
  Stores all measurable events (transactions).
  Supports KPIs such as total deposits, withdrawals, balances, and transaction counts.

Dimension Tables

- **Customer Dimension** → demographic analysis
- **Branch Dimension** → branch performance
- **City & Country Dimensions** → geographic insights
- **Account Type Dimension** → product segmentation
- **Date Dimension** → time-series reporting

Role in reporting

- Branch-wise performance dashboards
- Country-wise transaction analysis
- Customer segmentation

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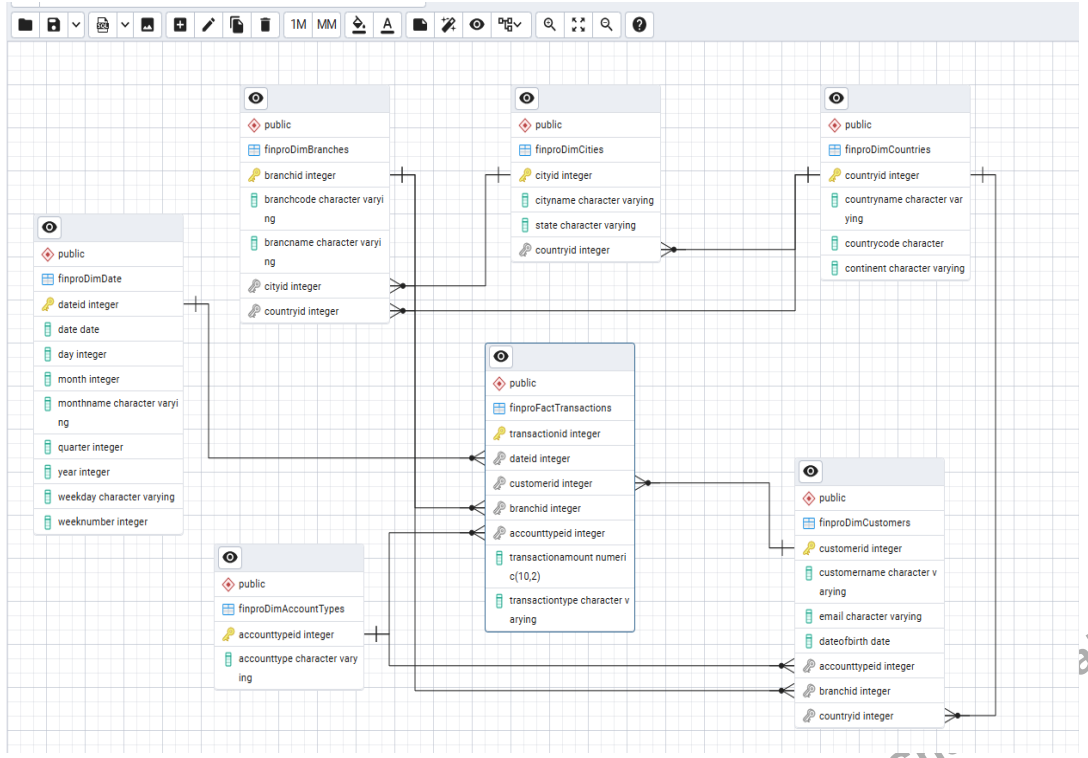
OLAP schema (3 of 3)

- The fact table sits at the center, connected to all dimensions.
- Each dimension provides a different analytical lens.
- The snowflake structure supports both granular and aggregated reporting.
- This design ensures fast, accurate, and scalable BI operations.



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Evaluation and interpretation

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Documenting SQL queries and scripts (1 of 7)

OLTP Scripts (Operational Transactions)

These scripts support day-to-day banking operations.

- **Create Tables (OLTP Layer)**
  - `CREATE TABLE finproDimCustomers (...)`
  - `CREATE TABLE finproFactTransactions (...)`
    (Purpose: Define operational entities and enforce constraints.)
- **Insert Data (OLTP Layer)**
  - `INSERT INTO finproDimCustomers (...) VALUES (...);`
  - `INSERT INTO finproFactTransactions (...) VALUES (...);`
    (Purpose: Load customer, branch, and transaction data.)

OLAP Scripts (Analytical Queries)

These scripts support reporting, dashboards, and insights.

- Aggregations, Joins, Grouping Sets, Rollup, Cube
  - Used for branch-wise totals, country-wise counts, and time-series analysis.

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Documenting SQL queries and scripts (2 of 7)

OLTP Queries (Insert, Update, Delete)

1. Insert Operation

```
INSERT INTO finproFactTransactions
(transactionid, dateid, customerid, branchid, accounttypeid,
transactionamount, transactiontype)
VALUES (10045, 20240115, 501, 12, 2, 850.00, 'Deposit');
```

Purpose: Adds a new transaction to the system.

2. Update Operation

```
UPDATE finproDimCustomers
SET email = 'updated_email@example.com'
WHERE customerid = 501;
```

Purpose: Corrects customer information.

3. Delete Operation

```
DELETE FROM finproFactTransactions
WHERE condition;
```

Purpose: Removes erroneous or duplicate transactions.

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Documenting SQL queries and scripts (3 of 7)

OLAP Queries (Aggregations, Joins, Rollup, Cube)

1. Branch-Wise Total Transaction Amount

```
SELECT b.branchname,
       SUM(f.transactionamount) AS total_amount
FROM finproFactTransactions f
JOIN finproDimBranches b ON f.branchid = b.branchid
GROUP BY b.branchname;
```

Purpose: Supports branch performance dashboards.

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Documenting SQL queries and scripts (4 of 7)

2. Multi-Level Aggregation Using GROUPING SETS

```
SELECT
    customerid,
    accountid,
    SUM(balance) AS total_balance
FROM finproDimCustomers
GROUP BY GROUPING SETS (
    (customerid, accountid),   -- detailed level
    (customerid),              -- subtotal by customer
    (accountid),               -- subtotal by account
    ()                         -- grand total
);
```

Purpose: Generates detailed, subtotal, and grand-total summaries for customer and account balances, supporting multi-dimensional financial analysis.

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Documenting SQL queries and scripts (5 of 7)

3. ROLLUP — Branch → Account Type → Total

```
SELECT
    customers.branchcode,
    customers.accounttype,
    SUM(facts.transactionamount) AS total_transaction
FROM finproFactTransactions AS facts
JOIN finproDimCustomers AS customers
    ON facts.accountid = customers.accountid
GROUP BY ROLLUP (customers.branchcode, customers.accounttype);
```

Purpose:

Generates hierarchical summaries:

- Branch + Account Type totals
- Branch-level subtotals
- Grand total

Supports multi-level reporting in dashboards.

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Documenting SQL queries and scripts (6 of 7)

4. CUBE — Year of Joining × Country × Balance

```
SELECT
    EXTRACT(YEAR FROM TO_DATE(customers.datejoined, 'MM/DD/YYYY')) AS year_of_joining,
    customers.country,
    customers.balance + COALESCE(SUM(facts.transactionamount), 0) AS closing_balance
FROM finproDimCustomers AS customers
LEFT JOIN finproFactTransactions AS facts
    ON customers.accountid = facts.accountid
GROUP BY CUBE (
    EXTRACT(YEAR FROM TO_DATE(customers.datejoined, 'MM/DD/YYYY')),
    customers.country,
    customers.balance
);
```

Purpose:

Creates all possible combinations of:

- Year
- Country
- Balance

Used for deep OLAP slicing/dicing and Tableau drill-downs.

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Documenting SQL queries and scripts (7 of 7)

5. MQT — Materialized View for Branch-Wise Totals

```
CREATE MATERIALIZED VIEW Total_transaction_per_branch AS
SELECT
    customers.branchcode,
    SUM(facts.transactionamount) AS total_transaction
FROM finproFactTransactions AS facts
JOIN finproDimCustomers AS customers
    ON facts.accountid = customers.accountid
GROUP BY customers.branchcode;
```

Querying the MQT

```
SELECT * FROM Total_transaction_per_branch;
```

Purpose:

- Improves performance by pre-computing branch-level totals.
- Used for fast dashboard refresh and repeated analytical queries.

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Inferences (1 of 4)

Inferences from OLTP Analysis

Customer Behavior Insights

- Negative balances highlight overdraft risk and potential financial stress.
- High positive balances identify premium-eligible, high-value customers.
- Irregular transaction patterns suggest possible data entry issues or unusual activity.

Transaction Frequency Patterns

- Frequent small withdrawals indicate cash-flow sensitivity.
- Large, infrequent deposits reflect salary or business-driven behavior.
- High withdrawals with low deposits may signal financial distress or fraud risk.

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Inferences (2 of 4)

Inferences from OLTP Analysis

Data Quality Observations

- Inconsistent emails and duplicate transactions show the need for stronger OLTP validation.

Actionable Takeaways

- Flag negative-balance accounts for follow-up.
- Target high-balance customers with tailored products.
- Strengthen validation rules to improve data accuracy.

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Inferences (3 of 4)

Inferences from OLAP Analysis

ROLLUP Insights (Branch → Account Type)

- Certain branches consistently outperform others.
- Specific account types drive most activity within each branch.
- Regional performance gaps highlight underperforming locations.

CUBE Insights (Year × Country × Balance)

- Clear year-over-year growth in some countries; declines in others.
- Geographic and joining-year combinations reveal high-value customer clusters.

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Inferences (4 of 4)

Inferences from OLAP Analysis

MQT Insights (Branch-Wise Totals)

- High-performing branches stand out clearly.
- Pre-computed totals improve dashboard performance and repeated analysis.
- Provides a reliable baseline for monthly and quarterly reviews.

Actionable Takeaways

- Invest in high-performing regions; investigate low-performing ones.
- Use geographic trends for targeted marketing and retention strategies.
- Leverage MQT outputs for faster, more consistent reporting.

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Data integration and security

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Data integration recommendations (1 of 3)

Areas Requiring Data Integration

Key Integration Points

- **Customer ↔ Branch ↔ Geography**
  Ensures accurate mapping of customers to branches, cities, and countries.
- **Transactions ↔ Customers**
  Required for closing balance, transaction frequency, and fraud analysis.
- **Date Dimension ↔ Fact Table**
  Enables time-series reporting (year, quarter, month, week).

Why Integration Matters

- Reduces data silos
- Ensures consistent reporting
- Supports OLAP queries and Tableau dashboards
- Improves data quality and governance

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Data integration recommendations (2 of 3)

Recommended Integration Patterns

1. ETL (Extract-Transform-Load)

- Ideal for scheduled batch loads
- Cleans and standardizes data before loading
- Fits FinPro's CSV → SQL → Tableau workflow

2. ELT (Extract-Load-Transform)

- Useful when raw data must be stored first
- Leverages SQL transformations inside the warehouse

3. CDC (Change Data Capture)

- Tracks inserts/updates/deletes
- Ensures near-real-time updates for transactions

4. API-Based Integration (Future-Ready)

- For connecting mobile banking, CRM, or fraud systems

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Data integration recommendations (3 of 3)

Alignment with FinPro Bank's Ecosystem

Why These Patterns Fit FinPro

- Current system uses **CSV ingestion + SQL warehouse** → ETL is ideal
- Transaction data changes frequently → CDC recommended
- Tableau dashboards require **clean, structured dimensions** → ELT/ETL hybrid works
- Future expansion (mobile app, CRM) → API integration readiness

Outcome

- Reliable data flow
- Faster reporting
- Scalable architecture for future banking needs

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Security documentation (1 of 3)

Role-Based Access Control (RBAC)

Allowed	Restricted
• SELECT on all dimension tables • SELECT + INSERT on fact table (for new transactions) • UPDATE on customer email/phone fields	• No DELETE on fact tables • No access to encrypted columns (DOB, email hash) • No privilege to alter schemas or create tables

Why RBAC Matters

- Protects sensitive data
- Prevents unauthorized changes
- Ensures auditability

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Security documentation (2 of 3)

Column-Level Encryption Guidelines

Sensitive Columns

- Customer email
- Date of birth
- Account number

Encryption Approach	fin_manager Access
•Use **AES_ENCRYPT / AES_DECRYPT** •Store encrypted values in separate columns •Keep encryption keys in a secure vault (not in SQL scripts)	•Can **view decrypted values only when necessary** •Cannot access encryption keys •Cannot modify encrypted columns directly

Outcome

- Protects PII
- Ensures compliance with banking security standards

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Security documentation (3 of 3)

Step-by-Step Procedure

1. **Create Role**
   ```
   CREATE ROLE fin_manager;
   ```
2. **Grant Read Access**
   ```
   GRANT SELECT ON ALL TABLES IN SCHEMA public TO fin_manager;
   ```
3. **Grant Limited Write Access**
   ```
   GRANT INSERT, UPDATE ON finproFactTransactions TO fin_manager;
   ```
4. **Restrict Sensitive Tables**
   ```
   REVOKE SELECT ON encrypted_columns FROM fin_manager;
   ```
5. **Test Permissions**
   Ensure fin_manager cannot delete or alter tables.

Outcome
•Controlled access •Reduced risk •Clear audit trail

Examples

- **Granted:** SELECT on customers, branches, countries
- **Restricted:** DELETE on fact tables, ALTER TABLE, access to encrypted fields

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Data reporting

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Report specifications (1 of 6)

Data Sources Imported into Tableau

- Account_Type.csv
  Contains account type categories used for customer segmentation and filtering.
- Branches.csv
  Includes branch codes, branch names, and geographic identifiers used for branch-wise reporting.
- Countries.csv
  Provides country names and codes used for country-wise transaction analysis and map visualizations.
- Customer_Data.xlsx
  Contains customer demographics, DOB (for age calculation), account type, and branch mapping.
- Transaction_Data.xlsx
  Stores transaction amounts, transaction types, dates, and customer IDs — the core source for closing balance and transaction count reports.
- US_Cities.csv
  Provides city-level geographic information used for branch/city mapping where applicable.

Relevance to Tableau Reports

- Customer-wise closing balance
- Branch-wise closing balance
- Country-wise transaction count
- Customer-wise average age
- Dashboard and story creation.

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Report specifications (2 of 6)

Filters & Calculated Fields

Key Filters

- Branch Filter → isolate performance by branch
- Country Filter → geographic segmentation
- Transaction Type Filter → deposits vs withdrawals
- Date Filter → year, quarter, month

Key Calculated Fields (These fields power all customer-wise, branch-wise, and country-wise dashboards.)

Closing Balance	[Balance] +
	IF [Transaction Type] = 'Deposit'
	  THEN [Transaction Amount]
	  ELSE -[Transaction Amount]
	END

Age	DATEDIFF('year', [Date of Birth], TODAY())
Rounded Average Age	ROUND(AVG([Age]))

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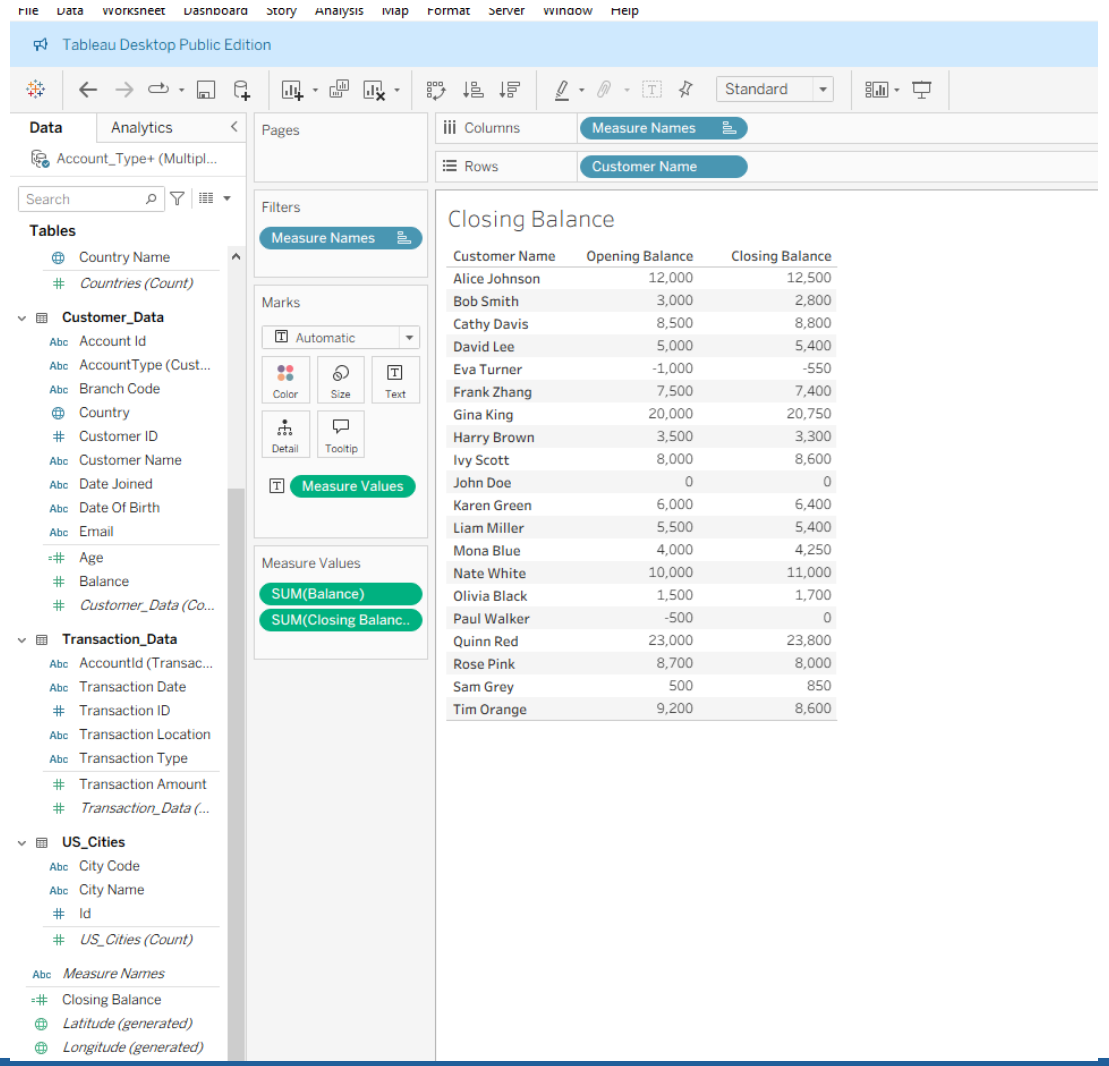
Report specifications (3 of 6)



Customer Name	Opening Balance	Closing Balance
Alice Johnson	12,000	12,500
Bob Smith	3,000	2,800
Cathy Davis	8,500	8,800
David Lee	5,000	5,400
Eva Turner	-1,000	-550
Frank Zhang	7,500	7,400
Gina King	20,000	20,750
Harry Brown	3,500	3,300
Ivy Scott	8,000	8,600
John Doe	0	0
Karen Green	6,000	6,400
Liam Miller	5,500	5,400
Mona Blue	4,000	4,250
Nate White	10,000	11,000
Olivia Black	1,500	1,700
Paul Walker	-500	0
Quinn Red	23,000	23,800
Rose Pink	8,700	8,000
Sam Grey	500	850
Tim Orange	9,200	8,600

Customer-Wise Closing Balance Report (Tableau)

Purpose

Shows each customer's final balance after all transactions — used for customer segmentation and risk assessment.

How It Was Built in Tableau

- Connected **fact + customer dimension**
- Created **Closing Balance** calculated field
- Used **SUM(Closing Balance)** as measure
- Placed **Customer Name** on Rows
- Sorted descending by balance
- Added **Branch** and **Country** filters
- Formatted as a **bar chart** for clarity

Relevance

Identifies high-value customers and accounts with negative balances.

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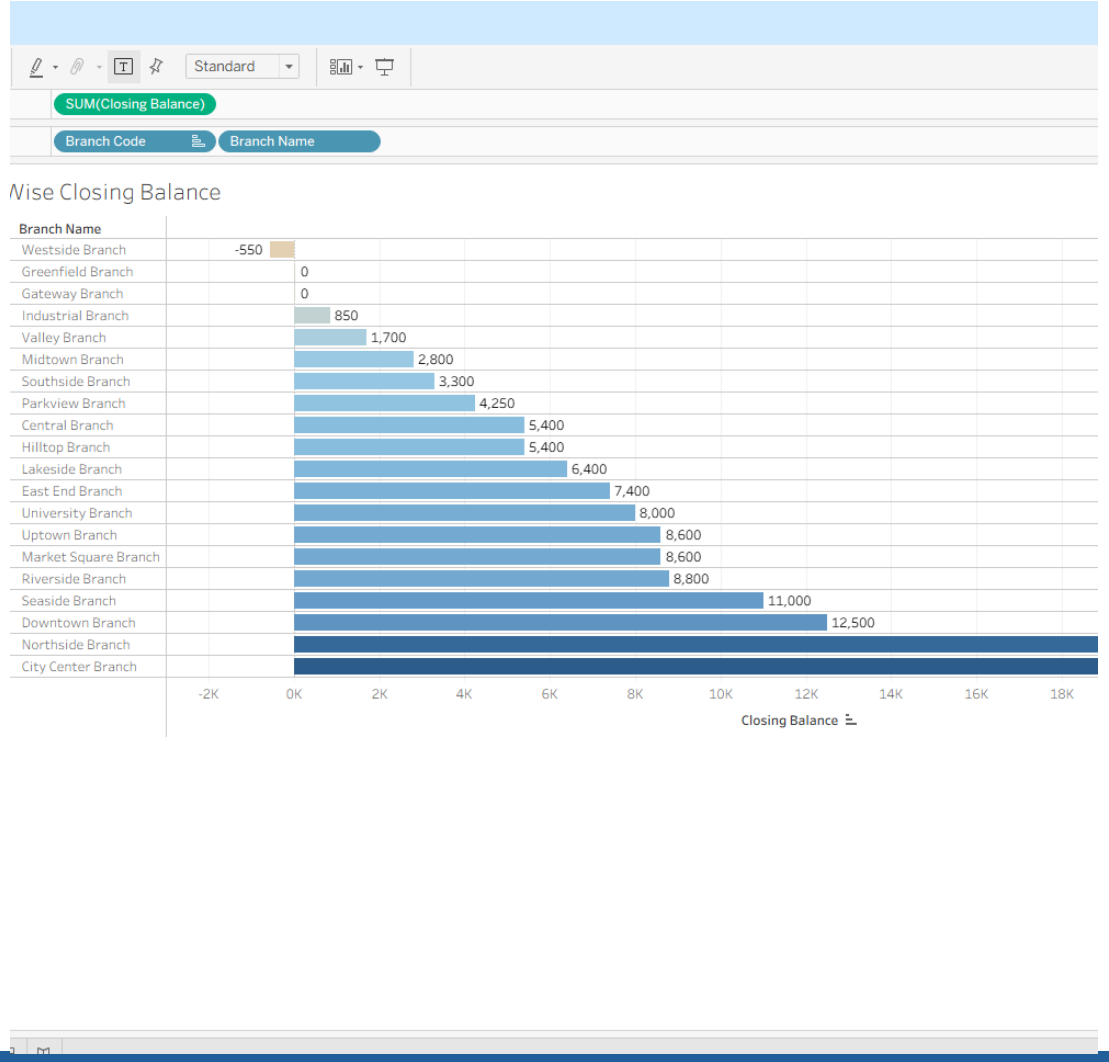
Report specifications (4 of 6)



Branch-Wise Closing Balance Report (Tableau)

Purpose

Shows total closing balance per branch — used for branch performance evaluation.

How It Was Built

- Aggregated **SUM(Closing Balance)** by **Branch Code**
- Used **Bar Chart** for comparison
- Added **Branch Filter**
- Applied color encoding for high vs low balance
- Used **Label Marks** for readability

Relevance

Supports regional performance reviews and resource allocation.

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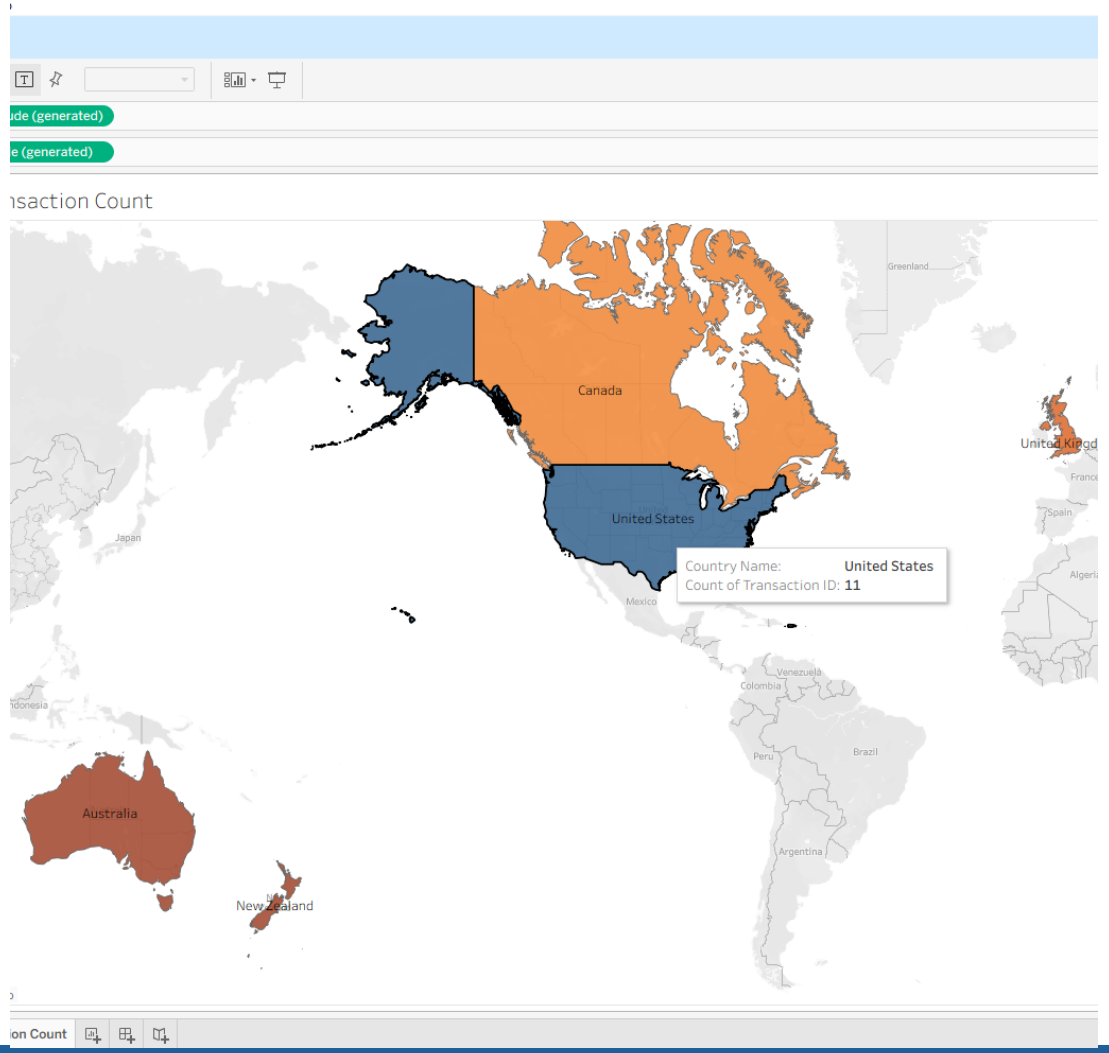
Report specifications (5 of 6)



Country-Wise Transaction Count Report (Tableau)

Purpose

Shows total number of transactions per country — used for geographic insights.

How It Was Built

- Joined **fact table** with **country dimension**
- Used **COUNT(TransactionID)**
- Built a **filled map** visualization
- Applied **Country Filter**
- Used color intensity to show transaction volume

Relevance

Highlights high-activity regions and potential growth markets.

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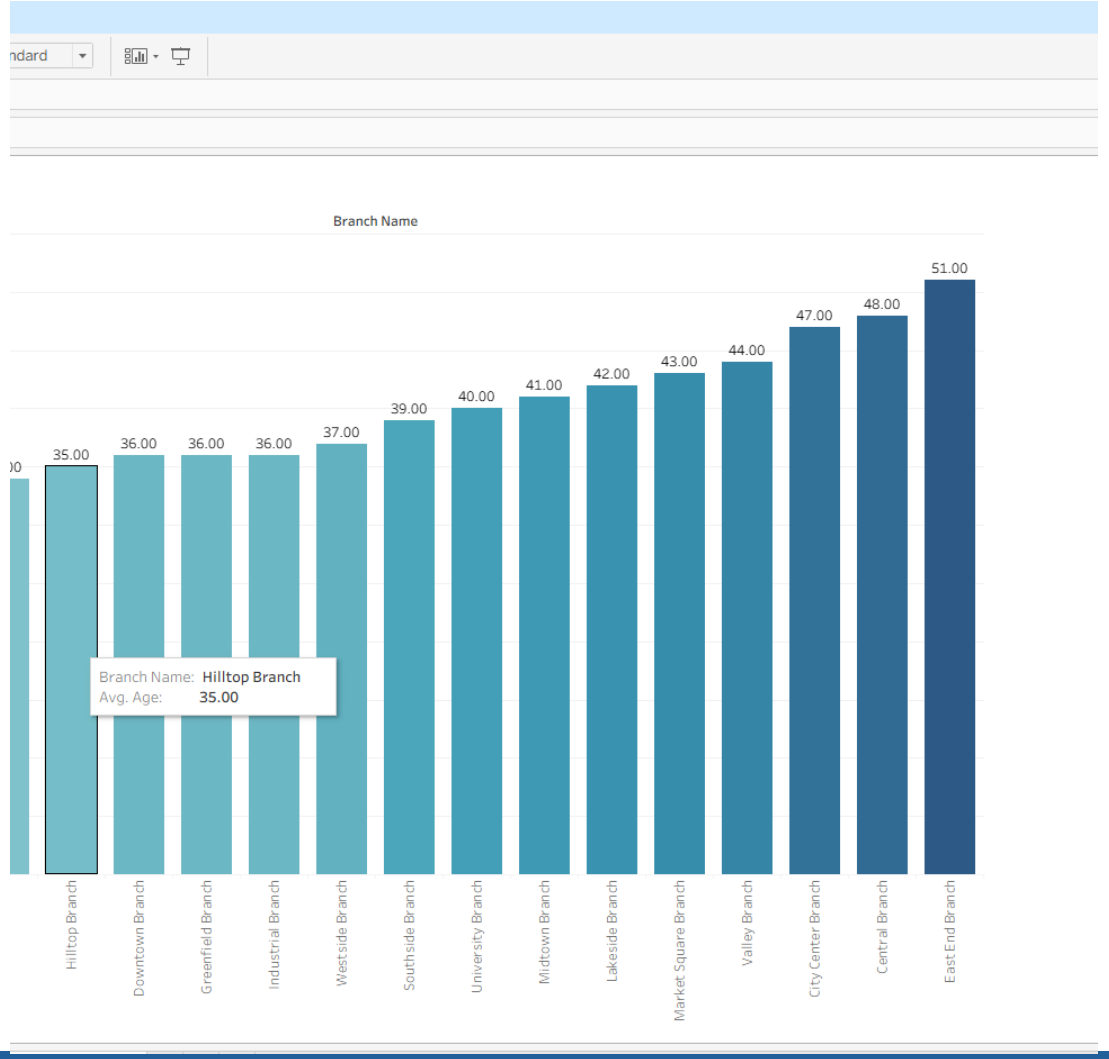
Report specifications (6 of 6)



Customer-Wise Average Age Report (Tableau)

Purpose

Shows average customer age per branch or country — used for demographic segmentation.

How It Was Built

- Created **Age** and **Rounded Average Age** fields
- Grouped by **Customer** or **Branch**
- Used **Bar Chart**
- Added filter for **Average Age**

Relevance

Supports targeted marketing and customer profiling.

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Performance analysis guidelines (1 of 3)

KPIs for Performance Analysis

Key KPIs Used

- *Total Transactions*
  COUNT(TransactionID)
- *Closing Balance*
  SUM(Closing Balance)
- Average Transaction Amount
  SUM(Transaction Amount) / COUNT(TransactionID)
- *Deposit vs Withdrawal Ratio*
  Deposits ÷ Withdrawals
- *Branch Performance Score*
  Weighted score using balance + transaction count
- Customer Average Age
  AVG(Age)

Role in Decision-Making
•Identifies high-value customers •Highlights strong and weak branches •Supports marketing and retention strategies •Helps detect anomalies or risk patterns

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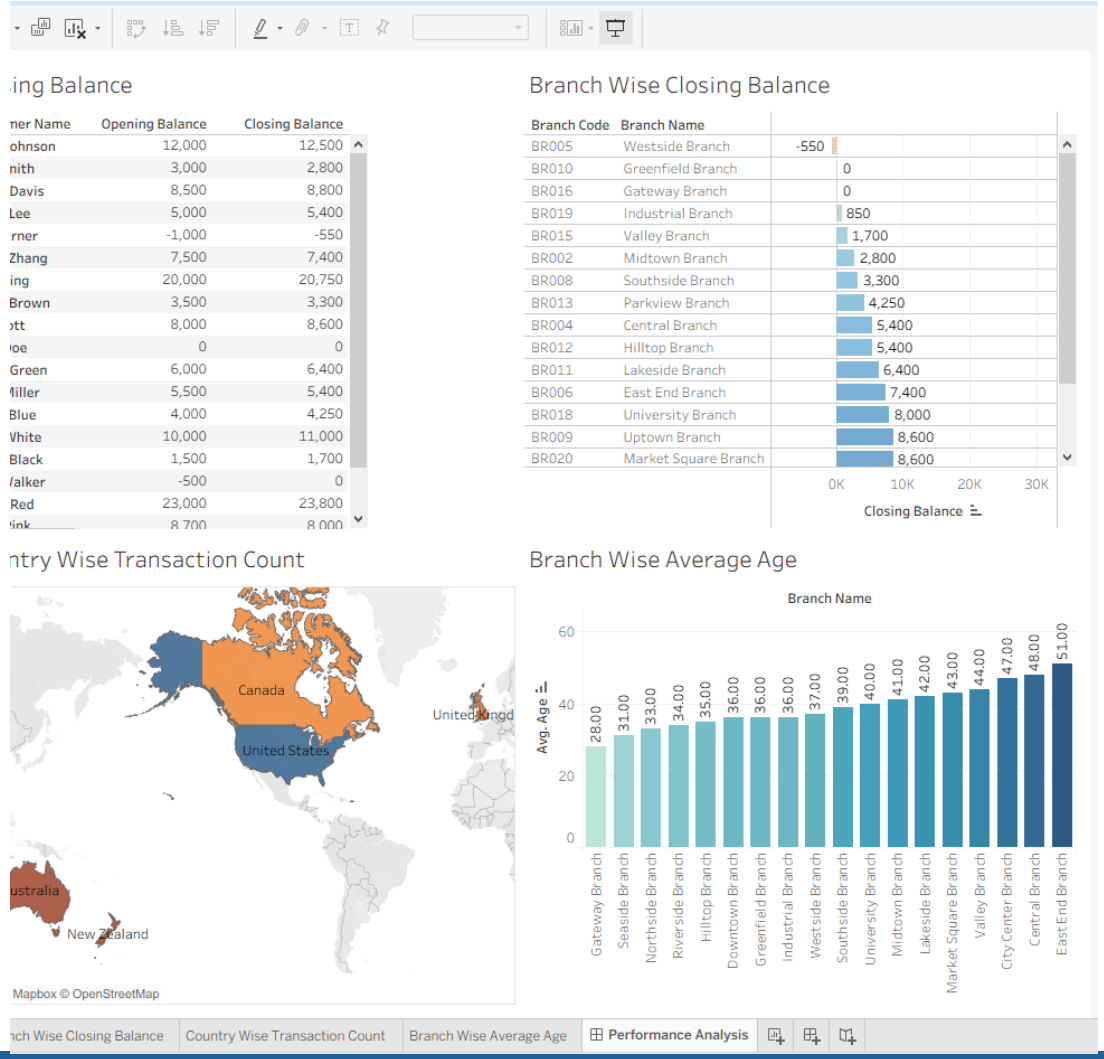
Performance analysis guidelines (2 of 3)



Dashboard, Story & Maintenance Steps

Dashboard Purpose

- Combines customer, branch, and country insights
- Enables interactive filtering and drill-down
- Provides a single view for decision-makers

Story Purpose

- Presents insights in a narrative sequence
- Highlights key findings from OLTP + OLAP analysis

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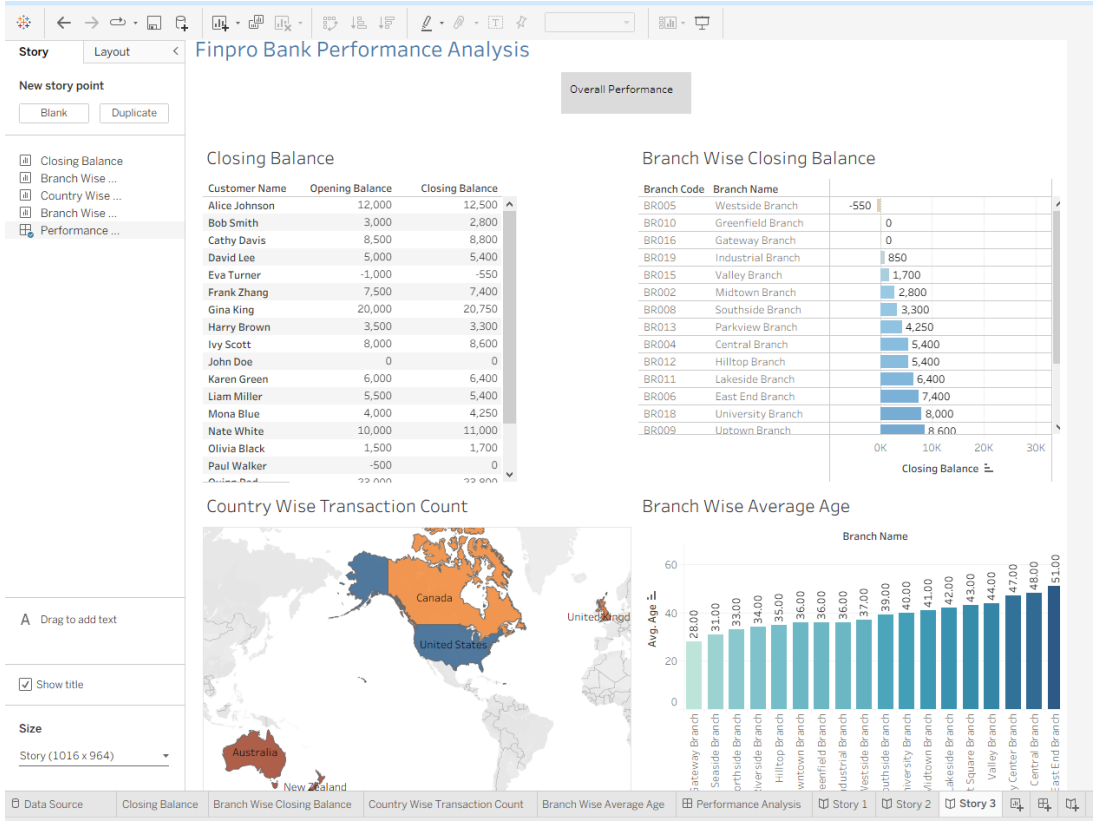
Performance analysis guidelines (3 of 3)



Maintenance Steps

- Refresh data sources (CSV → SQL → Tableau)
- Re-run ETL/ELT scripts
- Update calculated fields if schema changes
- Add/remove KPIs based on business needs
- Optimize performance by:
  - Using extracts
  - Removing unused fields
  - Aggregating data where possible

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General handover documentation

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Role and key responsibilities

Role Summary

As the **Data Engineer / Data Manager** for the FinPro Bank Capstone Project, I was responsible for designing, integrating, securing, and operationalizing the full data lifecycle — from ingestion to reporting.

Key Responsibilities

- Data Ingestion & Integration
- OLTP & OLAP Data Modeling
- SQL Development
- Tableau BI Development
- Data Quality & Governance
- Security & Access Control

Contribution to Project Goals

Delivered a secure, integrated, analytics-ready data environment enabling accurate reporting, faster insights, and reliable decision-making for FinPro Bank.

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System architecture diagram - i

High-Level Architecture (Data Engineer View)

Source Files (CSV / Excel)

↓ Data Cleaning & Validation

OLTP Layer (SQL Database)

Cleaned tables, PK/FK enforcement,
inserts/updates/deletes, operational queries

↓ Transformation & Aggregation

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System architecture diagram - ii

High-Level Architecture (Data Engineer View)

↓ Transformation & Aggregation

OLAP Layer (Analytical SQL)

ROLLUP, CUBE, GROUPING SETS,
Materialized Views, analytical joins

↓ BI Consumption

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System architecture diagram- iii

High-Level Architecture (Data Engineer View)

```
-------------------------------------------------------
                ↓ BI Consumption

          Tableau Reporting Layer
-------------------------------------------------------
Dashboards:
  - Customer Closing Balance
  - Branch Closing Balance
  - Country Transaction Count
  - Customer Average Age
Storyboards, KPIs, filters, calculated fields
-------------------------------------------------------
```

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Key challenges and resolutions - i

1. Schema Mismatches Between Files & SQL

Issue**:** Column names and data types differed between CSV/XLSX files and SQL schema.

Resolution: Standardized naming, fixed data types, aligned join keys (CustomerID, BranchCode, CountryID).

2. Data Quality Issues

Issue: Inconsistent email formats, missing DOB values, duplicate transactions, invalid dates.

Resolution: Cleaned data in Excel, applied validation rules, removed duplicates, corrected formats.

3. Tableau Join Problems

Issue: Wrong join types caused inflated or missing values.

Resolution: Rebuilt relationships using correct keys; validated row counts and aggregates.

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Key challenges and resolutions - ii

4. Closing Balance Calculation Errors

Issue: Deposits/withdrawals not aggregating correctly.

Resolution: Created a robust conditional calculated field and validated against sample transactions.

5. Performance Bottlenecks

Issue: Tableau dashboards slow due to large transaction data.

Resolution: Used extracts, removed unused fields, optimized filters, and aggregated data.

6. Security & RBAC Gaps

Issue: fin_manager role lacked clear permissions.

Resolution: Defined RBAC, restricted sensitive columns, assigned table-specific privileges.

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Insights and recommendations

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Insights and recommendations - i

1. Data Quality Drives Everything

FinPro’s source files (Customer, Transactions, Branches, Countries, Account Types, US Cities) required extensive cleaning — inconsistent emails, mismatched IDs, missing DOBs, and duplicate transactions. These issues directly affected SQL joins, OLAP aggregations, and Tableau dashboards.

Recommendation:

Establish a repeatable data-quality routine: validate schemas, standardize formats, enforce PK/FK relationships, and run duplicate checks before loading data into SQL or Tableau.

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Insights and recommendations - ii

2. Maintain a Clear OLTP → OLAP → Tableau Pipeline

The project relies on a structured flow:
Source Files → SQL OLTP → SQL OLAP → Tableau Dashboards
Your closing balance logic, ROLLUP/CUBE queries, and Tableau calculated fields all depend on consistent join keys and clean transformations.

Recommendation:
Document the integration logic, especially how closing balance is calculated, how OLAP outputs feed Tableau, and how relationships are defined. This prevents breakage when new data arrives or schemas evolve.

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Insights and recommendations - iii

3. Strengthen Security & Governance Early

The fin_manager role required careful RBAC design, column-level encryption guidelines, and table-specific privileges. These controls protect sensitive customer data and ensure compliance.

Recommendation:

Maintain a security playbook: define roles, restrict sensitive columns, outline encryption steps, and validate permissions regularly. This ensures safe, auditable access as the system grows.

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Summary

End-to-End Data Management for FinPro Bank

This capstone project demonstrated how a modern data pipeline transforms fragmented financial data into reliable, secure, and actionable insights. Through hands-on work across Excel, SQL, OLTP, OLAP, security, governance, and Tableau, the project showcased the full lifecycle of data management in a banking environment.

What the Project Achieved

- Cleaned and standardized raw data from multiple CSV/XLSX sources.
- Built OLTP structures and validated operational data using SQL.
- Designed an OLAP warehouse with rollups, cubes, and materialized views.
- Integrated data across all layers, ensuring consistency from source files to analytics.
- Implemented security and governance, including RBAC, encryption guidance, and controlled privileges.
- Delivered Tableau dashboards and stories that provide clear insights into customers, branches, transactions, and demographics.

Overall Impact

The project delivered a complete, secure, and analytics-ready data ecosystem — enabling FinPro Bank to make faster, data-driven decisions and demonstrating the full skill set of a modern Data Manager.

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Appendix

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Appendix

Excel Data Cleaning & Transformation Worksheets
Includes customer demographics, DOB corrections, balance categorization, age group segmentation, and validation checks performed during preprocessing.

CustomerID	CustomerName	Email	DateOfBirth	AccountId	AccountType	BranchCode	Country	DateJoined	Balance	CustomerAge	BalanceCategory	AgeGroup
101	Alice Johnson	alice.j@example.com	5/21/1990	ACCT-10002345	Savings	BR001	United States	1/15/2020	12000	35	High Balance	26-35
102	Bob Smith	bob.smith@xyz.com	8/15/1985	ACCT-10005678	Checking	BR002	United States	11/30/2019	3000	40	Low Balance	36-50
103	Cathy Davis	cathy.davis@example.com	11/3/1992	ACCT-10007891	Savings	BR003	United States	6/1/2021	8500	33	Medium Balance	26-35
104	David Lee	david.l@xyz.com	2/28/1978	ACCT-10001234	Savings	BR004	United States	3/20/2021	5000	48	Medium Balance	36-50
105	Eva Turner	No Email Provided	4/7/1989	ACCT-10003456	Checking	BR005	United States	7/19/2020	-1000	36	Low Balance	36-50
106	Frank Zhang	No Email Provided	12/13/1975	ACCT-10004567	Savings	BR006	United States	9/11/2018	7500	50	Medium Balance	36-50
107	Gina King	No Email Provided	3/20/1993	ACCT-10008901	Savings	BR007	Canada	1/9/2022	20000	32	High Balance	26-35
108	Harry Brown	No Email Provided	9/15/1987	ACCT-10006789	Checking	BR008	United States	8/21/2020	3500	38	Low Balance	36-50
109	Ivy Scott	No Email Provided		ACCT-10009876	Savings	BR009	Canada	11/14/2021	8000		Medium Balance	Missing DOB
110	John Doe	No Email Provided	1/1/1990	ACCT-10005432	Savings	BR010	United States	3/29/2022	0	36	Low Balance	36-50
111	Karen Green	No Email Provided	11/11/1984	ACCT-10007654	Savings	BR011	United States	12/5/2021	6000	41	Medium Balance	36-50
112	Liam Miller	No Email Provided	2/22/1991	ACCT-10004321	Checking	BR012	United Kingdom	5/25/2019	5500	35	Medium Balance	26-35
113	Mona Blue	No Email Provided		ACCT-10002134	Savings	BR013	United States	1/18/2022	4000		Low Balance	Missing DOB
114	Nate White	No Email Provided	10/10/1995	ACCT-10006543	Checking	BR014	Canada	6/30/2018	10000	30	High Balance	26-35
115	Olivia Black	No Email Provided	6/16/1982	ACCT-10003210	Savings	BR015	United States	7/19/2021	1500	43	Low Balance	36-50
116	Paul Walker	No Email Provided	7/5/1998	ACCT-10005478	Savings	BR016	Australia	9/25/2020	-500	27	Low Balance	26-35
117	Quinn Red	No Email Provided	4/22/1979	ACCT-10008765	Savings	BR017	United Kingdom	2/14/2019	23000	46	High Balance	36-50
118	Rose Pink	No Email Provided	8/8/1986	ACCT-10002301	Checking	BR018	New Zealand	3/9/2022	8700	39	Medium Balance	36-50
119	Sam Grey	No Email Provided	9/18/1990	ACCT-10007621	Checking	BR019	United Kingdom	4/4/2020	500	35	Low Balance	26-35
120	Tim Orange	No Email Provided	12/31/1983	ACCT-10003489	Savings	BR020	Canada	7/23/2019	9200	42	Medium Balance	36-50

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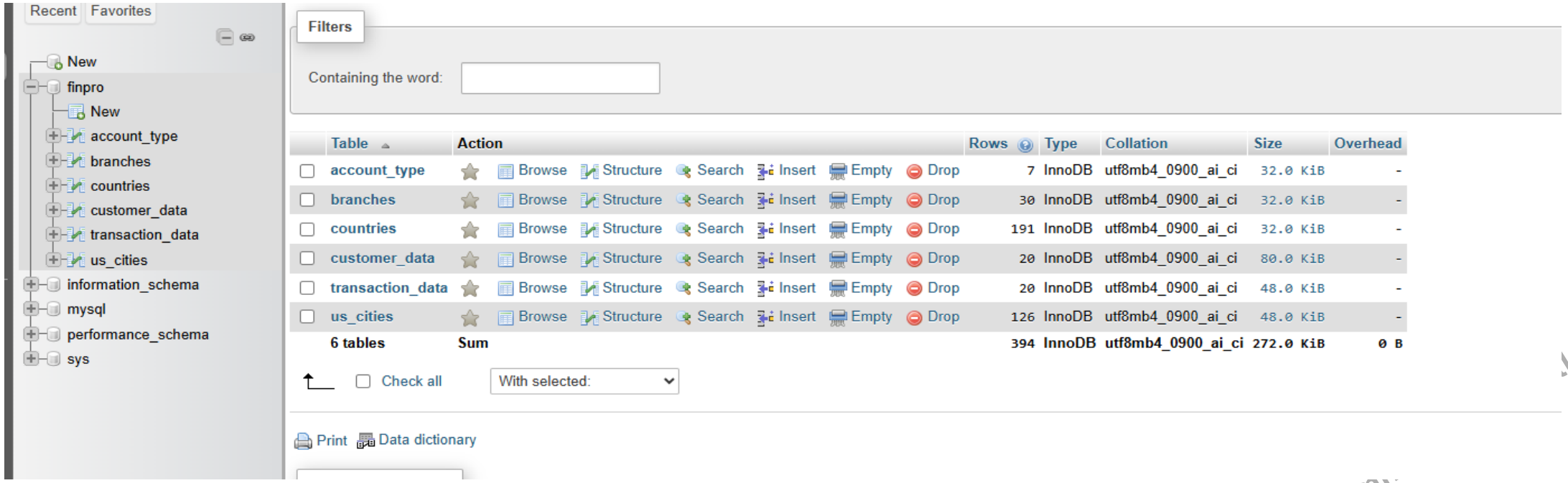
Appendix

Database Structure (phpMyAdmin View)
Shows the final SQL schema for the FinPro project, including:
account_type, branches, countries, customer_data, transaction_data, us_cities
Demonstrates successful data migration, table creation, and relational integrity.



Appendix

Tableau Links

Customer & Branch Performance Dashboard

Country-Wise Transaction Analysis

FinPro Storyboard

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